



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

TEST 2

SEMESTER III, SESSION 2020/2021

COURSE CODE : FSPC 0034
COURSE NAME : CHEMISTRY II
PROGRAMME : ASASI UTMSPACE
DURATION : 1 HOUR 15 MINUTES
DATE : 22 MARCH 2021

INSTRUCTION TO CANDIDATES:

1. DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.
2. THERE ARE **THREE (3)** QUESTIONS. ANSWER ALL QUESTIONS.

WARNING!

Students caught copying/cheating during the examination will be liable for disciplinary actions and UTMSPACE may recommend the student to be expelled from the study.

This booklet consists of (6) printed pages including this page.

QUESTION 1 (10 MARKS)

A buffer solution is prepared by mixing 100 cm³ aqueous NH₃ (0.1 mol dm⁻³) with 100 cm³ NH₄Cl (1.0 mol dm⁻³). Given $K_b \text{ NH}_3 = 1.74 \times 10^{-5} \text{ mol dm}^{-3}$

- a) Calculate the pH of the buffer solution. (4 marks)
- b) Determine the pH of the solution after the addition of HCl, (1 cm³, 0.1 mol dm⁻³). (6 marks)

QUESTION 2 (10 MARKS)

An electrochemical cell is formed when copper, Cu metal was dipped in an aqueous solution of 1.0 mol dm⁻³ CuSO₄ and connected to a standard hydrogen electrode at 25 °C, 1.0 M. The voltmeter reading of the cell is + 0.34 V.

- a) Draw a complete labelled diagram of the cell.
- b) Write the half-equations of the reaction at the anode and cathode.
- c) Write the overall equation of the cell.
- d) Write down the cell notation (IUPAC notation).
- e) State what can be observed at the electrode (at cathode and anode)

(4 + 2 + 1 + 1 + 2 marks)

$$1) \quad (a) \quad pOH = pK_b + \log\left(\frac{[NH_4^+]}{[NH_3]}\right)$$

$$= -\log(1.74 \times 10^{-5}) + \log\left(\frac{0.5}{0.05}\right)$$

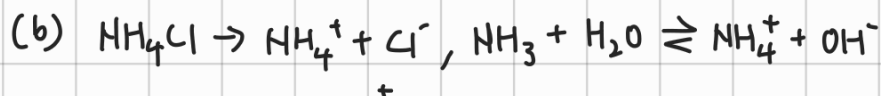
$$= 5.76$$

$$pH = 14 - 5.76$$

$$= 8.26$$

$$[NH_3] = \frac{(100)(0.1)}{200} = 0.05 \text{ mol dm}^{-3}$$

$$[NH_4^+] = \frac{100(1.0)}{200} = 0.5 \text{ mol dm}^{-3}$$



$$\text{mol of HCl added} = 0.001 \times 0.1 = 1 \times 10^{-4} \text{ mol}$$

$$\text{initial mol of } NH_3 = 0.1 \times 0.1 = 0.01 \text{ mol}$$

$$\text{initial mol of } NH_4^+ = 0.1 \times 1.0 = 0.1 \text{ mol}$$

	H^+	$+ NH_3$	$\rightarrow NH_4$ (increased)
init	1×10^{-4}	0.01	0.1
change	-1×10^{-4}	-1×10^{-4}	$+1 \times 10^{-4}$
equil	0	0.0099	0.1001

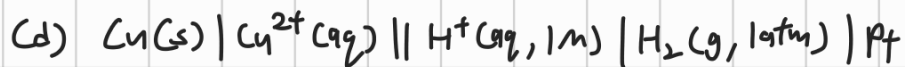
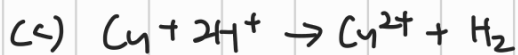
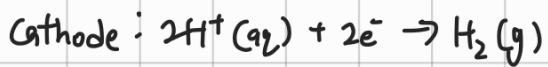
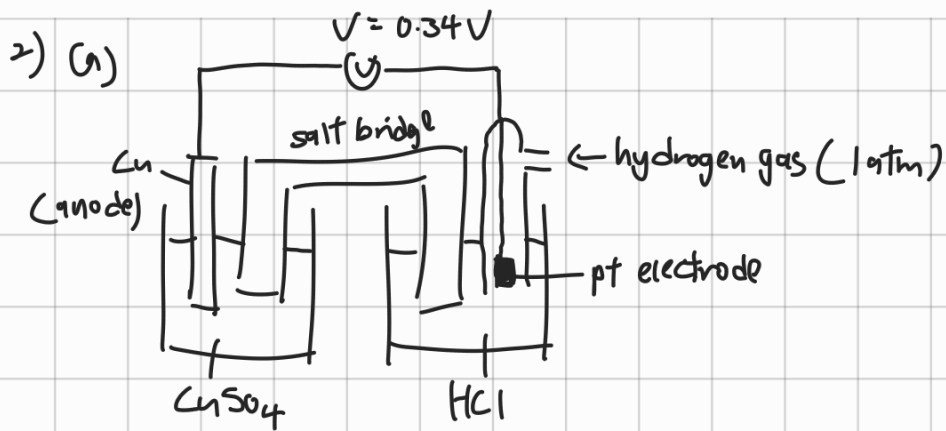
$$pOH = pK_b + \log\left(\frac{0.1001}{0.0099}\right)$$

$$= -\log(1.74 \times 10^{-5}) + 1.005$$

$$= 5.77$$

$$pH = 14 - 5.77$$

$$= 8.23$$

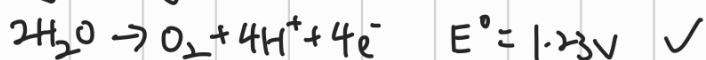
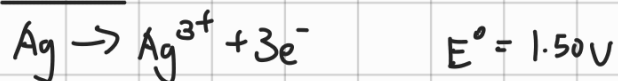


(e) Observation

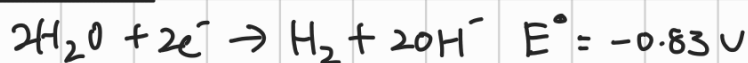
Anode : the copper electrode decrease in size

Cathode : colourless bubble gas produced

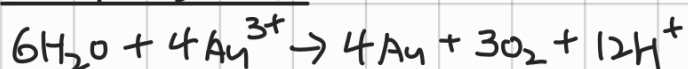
3) (a) Anode



Cathode



Overall Equation



(b) $m = eIt$

$$= \frac{AIt}{zF}$$

$$= \frac{197 \times 0.5 \times 6 \times 60 \times 60}{3 \times 96500}$$

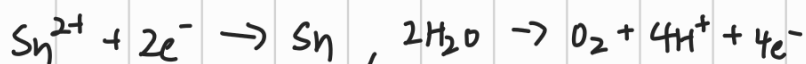
$$= 7.35 \text{ g of gold formed}$$

$$(c) \quad m = \frac{197 \times I \times 6 \times 60 \times 60}{3 \times 96500}$$

$$I = 0.68 \text{ A}$$

$$m = \frac{118.7 \times 0.68 \times 6 \times 60 \times 60}{2 \times 96500}$$

$$= 9.033 \text{ g}$$



Overall Equation



QUESTION 3 (10 MARKS)

Figure 1 shows Ali's attempt to electroplate an iron rod with gold (Au) by using gold chloride (AuCl_3) aqueous solution.

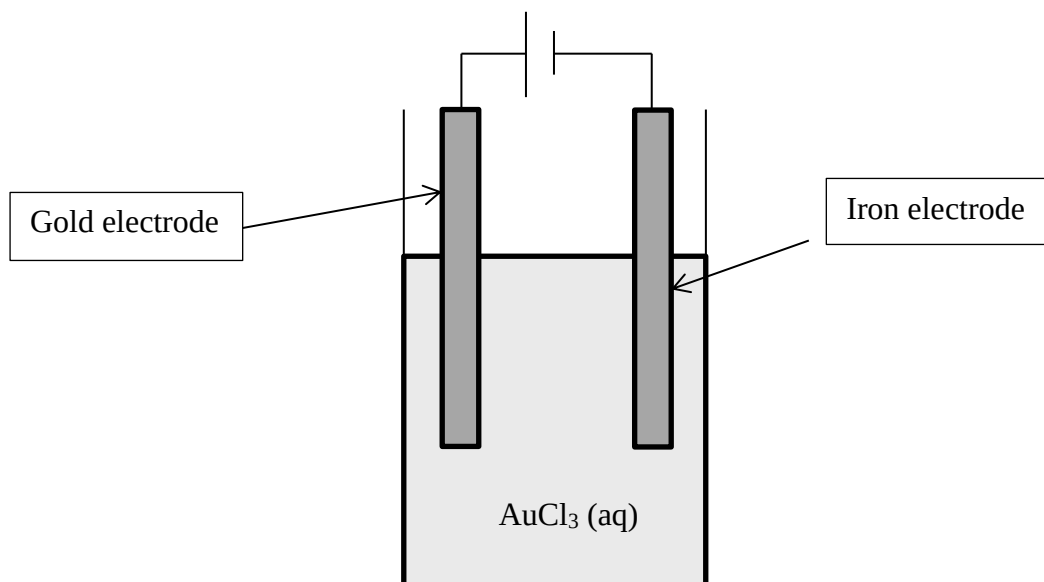


Figure 1

- Determine what will be the product at anode and the cathode by listing out all of the possible outcomes and choosing ONE most prominent product in each side. Write the overall equation for the redox reaction.
- In another experiment, Ali used a molten AuCl_3 instead of the aqueous AuCl_3 used above. If Ali applies 0.5 A for 6 hours, calculate the weight of the gold formed at the cathode.
- Ali cranked up the current supplied to an unknown value and collected 10 g of gold deposited on the iron rod in 6 hours. Now Ali applies the same setup but using molten tin chloride, SnCl_2 instead of molten AuCl_3 . Calculate the weight of tin deposited on the iron rod.

(4 + 3 + 3 marks)

Appendix 1 : THE PERIODIC TABLE

1	2	13	14	15	16	17	18
IA	IIA	IIIA	IVA	VA	VIA	VIIA	VIIIA
1 H 1.01	2 He 4.00	5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
3 Li 6.94	4 Be 9.01	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95
11 Na 22.99	12 Mg 24.31	31 Ga 69.72	32 Ge 72.61	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80
19 K 39.1	20 Ca 40.08	49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.6	53 I 126.9	54 Xe 131.29
37 Rb 85.47	38 Sr 87.62	81 Tl 204.4	82 Pb 207.2	83 Bi 209	84 Po (209)	85 At (210)	86 Rn (222)
55 Cs 132.9	56 Ba 137.3	89 Ac[^] (227)	90 Th 232.0	91 Pa (231)	92 U 238.0	93 Np (237)	94 Pu (244)
87 Fr (223)	88 Ra (226)	101 La[*] 138.9	102 Ce 140.1	103 Pr 140.9	104 Nd 144.2	105 Pm (145)	106 Sm 150.4
		107 Eu 152.0	108 Gd 157.3	109 Tb 158.9	110 Dy 162.5	111 Ho 164.9	112 Er 167.3
		113 Yb 173.0	114 Lu 175.0	115 Yt 176.9	116 La 178.9	117 Ce 180.9	118 Pr 182.9
		119 Sm 183.9	120 Eu 186.2	121 Gd 188.9	122 Tb 190.9	123 Dy 193.2	124 Ho 195.1
		125 Er 197.0	126 Tm 198.9	127 Yb 200.6	128 Lu 201.1	129 La 202.6	130 Ce 204.4
		131 Pr 205.9	132 Nd 207.2	133 Pm (209)	134 Sm (210)	135 Eu (211)	136 Gd (212)
		137 Tm 208.9	138 Yb 209.0	139 Lu 209.1	140 La 209.2	141 Ce 209.3	142 Pr 209.4
		143 Pr 209.5	144 Nd 209.6	145 Pm (209.7)	146 Sm (209.8)	147 Eu (209.9)	148 Gd (210.0)
		149 Tm 209.1	150 Yb 209.2	151 Lu 209.3	152 La 209.4	153 Ce 209.5	154 Pr 209.6
		155 Pr 209.7	156 Nd 209.8	157 Pm (209.9)	158 Sm (210.0)	159 Eu (210.1)	160 Gd (210.2)
		161 Tm 209.3	162 Yb 209.4	163 Lu 209.5	164 La 209.6	165 Ce 209.7	166 Pr 209.8
		167 Pr 209.9	168 Nd 210.0	169 Pm (210.1)	170 Sm (210.2)	171 Eu (210.3)	172 Gd (210.4)
		173 Tm 210.5	174 Yb 210.6	175 Lu 210.7	176 La 210.8	177 Ce 210.9	178 Pr 211.0
		179 Pr 211.1	180 Nd 211.2	181 Pm (211.3)	182 Sm (211.4)	183 Eu (211.5)	184 Gd (211.6)
		185 Tm 211.7	186 Yb 211.8	187 Lu 211.9	188 La 212.0	189 Ce 212.1	190 Pr 212.2
		191 Pr 212.3	192 Nd 212.4	193 Pm (212.5)	194 Sm (212.6)	195 Eu (212.7)	196 Gd (212.8)
		197 Tm 212.9	198 Yb 213.0	199 Lu 213.1	200 La 213.2	201 Ce 213.3	202 Pr 213.4
		203 Pr 213.5	204 Nd 213.6	205 Pm (213.7)	206 Sm (213.8)	207 Eu (213.9)	208 Gd (214.0)
		209 Tm 214.1	210 Yb 214.2	211 Lu 214.3	212 La 214.4	213 Ce 214.5	214 Pr 214.6
		215 Pr 214.7	216 Nd 214.8	217 Pm (214.9)	218 Sm (215.0)	219 Eu (215.1)	220 Gd (215.2)
		221 Tm 215.3	222 Yb 215.4	223 Lu 215.5	224 La 215.6	225 Ce 215.7	226 Pr 215.8
		227 Pr 215.9	228 Nd 216.0	229 Pm (216.1)	230 Sm (216.2)	231 Eu (216.3)	232 Gd (216.4)
		233 Tm 216.5	234 Yb 216.6	235 Lu 216.7	236 La 216.8	237 Ce 216.9	238 Pr 217.0
		239 Pr 217.1	240 Nd 217.2	241 Pm (217.3)	242 Sm (217.4)	243 Eu (217.5)	244 Gd (217.6)
		245 Tm 217.7	246 Yb 217.8	247 Lu 217.9	248 La 218.0	249 Ce 218.1	250 Pr 218.2
		251 Pr 218.3	252 Nd 218.4	253 Pm (218.5)	254 Sm (218.6)	255 Eu (218.7)	256 Gd (218.8)
		257 Tm 218.9	258 Yb 219.0	259 Lu 219.1	260 La 219.2	261 Ce 219.3	262 Pr 219.4
		263 Pr 219.5	264 Nd 219.6	265 Pm (219.7)	266 Sm (219.8)	267 Eu (219.9)	268 Gd (220.0)
		269 Tm 220.1	270 Yb 220.2	271 Lu 220.3	272 La 220.4	273 Ce 220.5	274 Pr 220.6
		275 Pr 220.7	276 Nd 220.8	277 Pm (220.9)	278 Sm (221.0)	279 Eu (221.1)	280 Gd (221.2)
		281 Tm 221.3	282 Yb 221.4	283 Lu 221.5	284 La 221.6	285 Ce 221.7	286 Pr 221.8
		287 Pr 221.9	288 Nd 222.0	289 Pm (222.1)	290 Sm (222.2)	291 Eu (222.3)	292 Gd (222.4)
		293 Tm 222.5	294 Yb 222.6	295 Lu 222.7	296 La 222.8	297 Ce 222.9	298 Pr 223.0
		299 Pr 223.1	300 Nd 223.2	301 Pm (223.3)	302 Sm (223.4)	303 Eu (223.5)	304 Gd (223.6)
		305 Tm 223.7	306 Yb 223.8	307 Lu 223.9	308 La 224.0	309 Ce 224.1	310 Pr 224.2
		311 Pr 224.3	312 Nd 224.4	313 Pm (224.5)	314 Sm (224.6)	315 Eu (224.7)	316 Gd (224.8)
		317 Tm 224.9	318 Yb 225.0	319 Lu 225.1	320 La 225.2	321 Ce 225.3	322 Pr 225.4
		323 Pr 225.5	324 Nd 225.6	325 Pm (225.7)	326 Sm (225.8)	327 Eu (225.9)	328 Gd (226.0)
		329 Tm 226.1	330 Yb 226.2	331 Lu 226.3	332 La 226.4	333 Ce 226.5	334 Pr 226.6
		335 Pr 226.7	336 Nd 226.8	337 Pm (226.9)	338 Sm (227.0)	339 Eu (227.1)	340 Gd (227.2)
		341 Tm 227.3	342 Yb 227.4	343 Lu 227.5	344 La 227.6	345 Ce 227.7	346 Pr 227.8
		347 Pr 227.9	348 Nd 228.0	349 Pm (228.1)	350 Sm (228.2)	351 Eu (228.3)	352 Gd (228.4)
		353 Tm 228.5	354 Yb 228.6	355 Lu 228.7	356 La 228.8	357 Ce 228.9	358 Pr 229.0
		359 Pr 229.1	360 Nd 229.2	361 Pm (229.3)	362 Sm (229.4)	363 Eu (229.5)	364 Gd (229.6)
		365 Tm 229.7	366 Yb 229.8	367 Lu 229.9	368 La 230.0	369 Ce 230.1	370 Pr 230.2
		371 Pr 230.3	372 Nd 230.4	373 Pm (230.5)	374 Sm (230.6)	375 Eu (230.7)	376 Gd (230.8)
		377 Tm 230.9	378 Yb 231.0	379 Lu 231.1	380 La 231.2	381 Ce 231.3	382 Pr 231.4
		383 Pr 231.5	384 Nd 231.6	385 Pm (231.7)	386 Sm (231.8)	387 Eu (231.9)	388 Gd (232.0)
		389 Tm 232.1	390 Yb 232.2	391 Lu 232.3	392 La 232.4	393 Ce 232.5	394 Pr 232.6
		395 Pr 232.7	396 Nd 232.8	397 Pm (232.9)	398 Sm (233.0)	399 Eu (233.1)	400 Gd (233.2)
		401 Tm 233.3	402 Yb 233.4	403 Lu 233.5	404 La 233.6	405 Ce 233.7	406 Pr 233.8
		407 Pr 233.9	408 Nd 234.0	409 Pm (234.1)	410 Sm (234.2)	411 Eu (234.3)	412 Gd (234.4)
		413 Tm 234.5	414 Yb 234.6	415 Lu 234.7	416 La 234.8	417 Ce 234.9	418 Pr 235.0
		419 Pr 235.1	420 Nd 235.2	421 Pm (235.3)	422 Sm (235.4)	423 Eu (235.5)	424 Gd (235.6)
		425 Tm 235.7	426 Yb 235.8	427 Lu 235.9	428 La 236.0	429 Ce 236.1	430 Pr 236.2
		431 Pr 236.3	432 Nd 236.4	433 Pm (236.5)	434 Sm (236.6)	435 Eu (236.7)	436 Gd (236.8)
		437 Tm 236.9	438 Yb 237.0	439 Lu 237.1	440 La 237.2	441 Ce 237.3	442 Pr 237.4
		443 Pr 237.5	444 Nd 237.6	445 Pm (237.7)	446 Sm (237.8)	447 Eu (237.9)	448 Gd (238.0)
		449 Tm 238.1	450 Yb 238.2	451 Lu 238.3	452 La 238.4	453 Ce 238.5	454 Pr 238.6
		455 Pr 238.7	456 Nd 238.8	457 Pm (238.9)	458 Sm (239.0)	459 Eu (239.1)	460 Gd (239.2)
		461 Tm 239.3	462 Yb 239.4	463 Lu 239.5	464 La 239.6	465 Ce 239.7	466 Pr 239.8
		467 Pr 239.9	468 Nd 240.0	469 Pm (240.1)	470 Sm (240.2)	471 Eu (240.3)	472 Gd (240.4)
		473 Tm 240.5	474 Yb 240.6	475 Lu 240.7	476 La 240.8	477 Ce 240.9	478 Pr 241.0
		479 Pr 241.1	480 Nd 241.2	481 Pm (241.3)	482 Sm (241.4)	483 Eu (241.5)	484 Gd (241.6)
		485 Tm 241.7	486 Yb 241.8	487 Lu 241.9	488 La 242.0	489 Ce 242.1	490 Pr 242.2
		491 Pr 242.3	492 Nd 242.4	493 Pm (242.5)	494 Sm (242.6)	495 Eu (242.7)	496 Gd (242.8)
		497 Tm 242.9	498 Yb 243.0	499 Lu 243.1	500 La 243.2	501 Ce 243.3	502 Pr 243.4
		503 Pr 243.5	504 Nd 243.6	505 Pm (243.7)	506 Sm (243.8)	507 Eu (243.9)	508 Gd (244.0)
		509 Tm 244.1	510 Yb 244.2	511 Lu 244.3	512 La 244.4	513 Ce 244.5	514 Pr 244.6
		515 Pr 244.7	516 Nd 244.8	517 Pm (244.9)	518 Sm (245.0)	519 Eu (245.1)	520 Gd (245.2)
		521 Tm 245.3	522 Yb 245.4	523 Lu 245.5	524 La 245.6	525 Ce 245.7	526 Pr 245.8
		527 Pr 245.9	528 Nd 246.0	529 Pm (246.1)	530 Sm (246.2		

Appendix 2 : LIST OF SELECTED CONSTANT VALUES

Speed of light in a vacuum	c	=	$3.0 \times 10^8 \text{ m s}^{-1}$
Avogadro's number	N_A	=	$6.02 \times 10^{23} \text{ mol}^{-1}$
Faraday constant	F	=	$9.65 \times 10^4 \text{ C mol}^{-1}$
Planck constant	h	=	$6.6256 \times 10^{-34} \text{ J s}$
Reduced Planck constant	$\hbar$	=	$1.054 \times 10^{-34} \text{ J s}$
Rydberg constant	R_H	=	$1.097 \times 10^7 \text{ m}^{-1}$ = $2.18 \times 10^{-14} \text{ J}$
Molar of gases constant	R	=	$8.314 \text{ J K}^{-1} \text{ mol}^{-1}$ = $0.08206 \text{ L atm mol}^{-1} \text{ K}^{-1}$
Boltzmann constant	k	=	$1.3807 \times 10^{-27} \text{ J K}^{-1}$
Mass of proton	M_p	=	$1.672 \times 10^{-27} \text{ kg}$
Electronic Bohr magneton	m_B	=	$9.2741 \times 10^{-24} \text{ J T}^{-1}$
Nuclear Bohr magneton	b_N	=	$5.05 \times 10^{-27} \text{ J T}^{-1}$
Vapour pressure of water	P_{water}	=	23.8 torr
Electron charge	e^-	=	$1.602 \times 10^{-19} \text{ C}$

UNIT AND CONVERSION FACTOR

Energy	$1 \text{ J} = 1 \text{ kg m}^2 \text{ s}^{-2} = 1 \text{ N m} = 10^7 \text{ erg}$ $1 \text{ calorie} = 4.184 \text{ Joule}$ $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$ $1 \text{ amu} = 1.66 \times 10^{-27} \text{ kg}$
Pressure	$1 \text{ atm} = 760 \text{ mm Hg} = 760 \text{ torr} = 101.325 \text{ kPa} = 101325 \text{ N m}^{-2}$

Appendix 3 : STANDARD REDUCTION POTENTIAL (E°_{Red})

$\text{PbO}_2(\text{s}) + \text{SO}_4^{2-}(\text{aq}) + 4\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{PbSO}_4(\text{s}) + 2\text{H}_2\text{O}(\ell)$	+1.69
$\text{MnO}_4^-(\text{aq}) + 8\text{H}^+(\text{aq}) + 5\text{e}^- \rightarrow$	$\text{Mn}^{2+}(\text{aq}) + 4\text{H}_2\text{O}(\ell)$	+1.51
$\text{Au}^{3+}(\text{aq}) + 3\text{e}^- \rightarrow$	$\text{Au}(\text{s})$	+1.50
$\text{ClO}_4^-(\text{aq}) + 8\text{H}^+(\text{aq}) + 8\text{e}^- \rightarrow$	$\text{Cl}^-(\text{aq}) + 4\text{H}_2\text{O}(\ell)$	+1.39
$\text{Cl}_2(\text{g}) + 2\text{e}^- \rightarrow$	$2\text{Cl}^-(\text{aq})$	+1.36
$\text{Cr}_2\text{O}_7^{2-}(\text{aq}) + 14\text{H}^+(\text{aq}) + 6\text{e}^- \rightarrow$	$2\text{Cr}^{3+}(\text{aq}) + 7\text{H}_2\text{O}(\ell)$	+1.33
$2\text{HNO}_3(\text{aq}) + 4\text{H}^+(\text{aq}) + 4\text{e}^- \rightarrow$	$\text{N}_2\text{O}(\text{g}) + 3\text{H}_2\text{O}(\ell)$	+1.30
$\text{O}_2(\text{g}) + 4\text{H}^+(\text{aq}) + 4\text{e}^- \rightarrow$	$2\text{H}_2\text{O}(\ell)$	+1.23
$\text{MnO}_2(\text{s}) + 4\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{Mn}^{2+}(\text{aq}) + 2\text{H}_2\text{O}(\ell)$	+1.22
$\text{Br}_2(\text{aq}) + 2\text{e}^- \rightarrow$	$2\text{Br}^-(\text{aq})$	+1.07
$\text{Hg}_2^{2+}(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{Hg}(\ell)$	+0.85
$\text{ClO}^-(\text{aq}) + \text{H}_2\text{O}(\ell) + 2\text{e}^- \rightarrow$	$\text{Cl}^-(\text{aq}) + 2\text{OH}^-(\text{aq})$	+0.84
$\text{Ag}^+(\text{aq}) + \text{e}^- \rightarrow$	$\text{Ag}(\text{s})$	+0.80
$\text{NO}_3^-(\text{aq}) + 2\text{H}^+(\text{aq}) + \text{e}^- \rightarrow$	$\text{NO}_2(\text{g}) + \text{H}_2\text{O}(\ell)$	+0.80
$\text{Fe}^{3+}(\text{aq}) + \text{e}^- \rightarrow$	$\text{Fe}^{2+}(\text{aq})$	+0.77
$\text{O}_2(\text{g}) + 2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{H}_2\text{O}_2(\ell)$	+0.70
$\text{I}_2(\text{s}) + 2\text{e}^- \rightarrow$	$2\text{I}^-(\text{aq})$	+0.54
$\text{O}_2(\text{g}) + 2\text{H}_2\text{O}(\ell) + 4\text{e}^- \rightarrow$	$4\text{OH}^-(\text{aq})$	+0.40
$\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{Cu}(\text{s})$	+0.34
$\text{SO}_4^{2-}(\text{aq}) + 4\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{H}_2\text{SO}_3(\text{aq}) + \text{H}_2\text{O}(\ell)$	+0.17
$\text{Sn}^{4+}(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{Sn}^{2+}(\text{aq})$	+0.15
$\text{S}(\text{s}) + 2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{H}_2\text{S}(\text{aq})$	+0.14
$\text{AgBr}(\text{s}) + \text{e}^- \rightarrow$	$\text{Ag}(\text{s}) + \text{Br}^-(\text{aq})$	+0.07
$2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{H}_{2(\text{g})}$	0.00
$\text{Pb}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{Pb}(\text{s})$	-0.13
$\text{Sn}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{Sn}(\text{s})$	-0.14
$\text{AgI}(\text{s}) + \text{e}^- \rightarrow$	$\text{Ag}(\text{s}) + \text{I}^-(\text{aq})$	-0.15
$\text{Ni}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{Ni}(\text{s})$	-0.26
$\text{Co}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{Co}(\text{s})$	-0.28
$\text{PbSO}_4(\text{s}) + 2\text{e}^- \rightarrow$	$\text{Pb}(\text{s}) + \text{SO}_4^{2-}(\text{aq})$	-0.36
$\text{Se}(\text{s}) + 2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{H}_2\text{Se}(\text{aq})$	-0.40
$\text{Cd}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{Cd}(\text{s})$	-0.40
$\text{Cr}^{3+}(\text{aq}) + \text{e}^- \rightarrow$	$\text{Cr}^{2+}(\text{aq})$	-0.41
$\text{Fe}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{Fe}(\text{s})$	-0.45
$\text{NO}_2^-(\text{aq}) + \text{H}_2\text{O}(\ell) + \text{e}^- \rightarrow$	$\text{NO}(\text{g}) + 2\text{OH}^-(\text{aq})$	-0.46
$\text{Ag}_2\text{S}(\text{s}) + 2\text{e}^- \rightarrow$	$2\text{Ag}(\text{s}) + \text{S}^{2-}(\text{aq})$	-0.69
$\text{Zn}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{Zn}(\text{s})$	-0.76
$2\text{H}_2\text{O}(\ell) + 2\text{e}^- \rightarrow$	$\text{H}_2(\text{g}) + 2\text{OH}^-(\text{aq})$	-0.83
$\text{Cr}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{Cr}(\text{s})$	-0.91
$\text{Se}(\text{s}) + 2\text{e}^- \rightarrow$	$\text{Se}^{2-}(\text{aq})$	-0.92
$\text{SO}_4^{2-}(\text{aq}) + \text{H}_2\text{O}(\ell) + 2\text{e}^- \rightarrow$	$\text{SO}_3^{2-}(\text{aq}) + 2\text{OH}^-(\text{aq})$	-0.93
$\text{Al}^{3+}(\text{aq}) + 3\text{e}^- \rightarrow$	$\text{Al}(\text{s})$	-1.66
$\text{Mg}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{Mg}(\text{s})$	-2.37
$\text{Na}^+(\text{aq}) + \text{e}^- \rightarrow$	$\text{Na}(\text{s})$	-2.71
$\text{Ca}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{Ca}(\text{s})$	-2.87
$\text{Ba}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{Ba}(\text{s})$	-2.91
$\text{Li}^+(\text{aq}) + \text{e}^- \rightarrow$	$\text{Li}(\text{s})$	-3.04